

You can check [this video](#) on max/min vs arg max/arg min.

Question 1: MDPs

Recall a Markov Decision Process (MDP) represents a scenario where actions are non-deterministic. Consider the following minor gridworld example:

	1	2	3
1	20		5
2			
3	-20		

Cells are indexed by (row, column). Cells (1, 1), (3, 1) and (1, 3) are terminal cells, and have rewards of 20, -20 and 5, respectively. For all other cells, there is an reward of -1.¹ The agent starts at cell (3, 3). The agent can move in north, east, south and west directions or stay where it is. The actions are stochastic and have the following outcomes:

- If go north, 70% of the time the agent will go north, 30% will go west.
- If go east, 60% of the time the agent will go east, 40% of the time will go south.
- If go south, 80% of the time the agent will go south, 20% of the time will go east.
- If go west, 70% of the time the agent will go west, 30% of the time will go north.
- If stay in current cell, 100% of the time will stay.

If an action causes the agent to travel into a wall, the agent will stay in their current cell.

- (a) Construct a MDP for this instance of gridworld. Recall a MDP consists of S (set of states), s_0 (starting state), possible terminal states, A (set of actions), T (transition model/function) and R (reward function), so all of these have to be specified. To help, consider constructing the MDP in steps:
- What are the states in this gridworld?
 - What are the starting and terminal states?
 - What are the set of actions?

¹Note this is taking a framework where rewards are given to states (i.e., $R(s)$) and not as transitions (i.e., $R(s, a, s')$) as done in the book. As we have discussed in class, both are standard conventions and they can be reduced to each other without loss of generality.

- iv. With cell (3, 3) as the current state s , construct the transition model/function for all actions and subsequent states, i.e., $T(s, a, s')$, where a are all possible actions for (3, 3) and s' are all possible successor states from (3, 3).

Recall, the definition of the transition function T is,

$$T : S \times A \times S \rightarrow \mathbb{R}, (s, a, s') \mapsto P(s'|a \wedge s)$$

- v. What is the reward function?

- (b) Using the MDP built, consider the following sequences of states that our agent progressed through. For each sequence, compute the *additive reward* and *discounted reward* (with a decay factor γ of 0.5). Take note of the differences between the two approaches.

(a) (3, 3), (2, 3), (2, 2), (2, 1), (1, 1)

(b) (3, 3), (3, 2), (2, 2), (2, 3), ... (repeat)

Here, we assume the cumulative reward for a set of states $s_0, s_1, \dots, s_t$ is $\sum_{i=0}^t R(s_i)$ (with discounting if you like). Note that another convention, as UCB slides or the textbook, is to use a reward function $R(s, a, s')$ on transitions, i.e., the reward given as you move from state s to state s' taking action a . In this scenario, you cannot have a reward for s_0 , as we don't have any state it has come from previously.

Question 2: Value iteration.....

- (a) Consider the MDP from Question 1, with a decay factor $\gamma = 1$.
- i. Using value iteration, compute the utilities for each state after two iterations. For this question take the value iteration equation:

$$U_{i+1}(s) = R(s) + \gamma \left(\max_a \sum_{s' \in S} T(s, a, s') U_i(s') \right).$$

Initially, we set $U_0(s) = 0$ for all states $s \in S$. Remember terminal states do not get more rewards once reached.

- ii. Repeat and determine the values after three iterations.
- (b) Consider the utilities after two iterations of value iteration from question a (we typically only perform this when values have converged, but for this question, do this after 2 iterations). Using the Maximum Expected Utility principle, find the best action (i.e., policy) for each state.

Recall, the equation to extract the optimal policy for a state s is:

$$\pi^*(s) = \operatorname{argmax}_a \sum_{s' \in S} T(s, a, s') U(s')$$

- (c) Consider the following policy for the MDP of question 1, with a decay factor γ of 1:

- $\pi((2, 1)) = \text{west}$.
- $\pi((2, 2)) = \text{west}$.
- $\pi((2, 3)) = \text{north}$.
- $\pi((3, 2)) = \text{north}$.
- $\pi((3, 3)) = \text{east}$.
- $\pi((1, 2)) = \text{south}$.

Evaluate this policy, i.e., compute the utility for each state, for one and two iterations (we typically can solve it as a system of simultaneous equations but for this question we use an iterative approach).

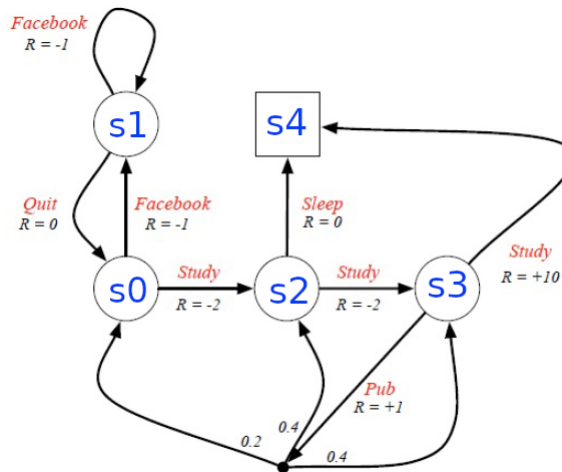
- (d) For any possible gridworld MDP, is the stay action ever part of an optimal policy? If yes, design a gridworld that demonstrates it, or if no prove that it is impossible. Hint: consider the reward function for non-terminal cells.

Question 3: More on MDP modeling

When defining an MDP, one can choose to represent the rewards per state, via a function $R(s)$ (i.e., “the immediate reward received at state s ”) or assign rewards to each transition via a function $R(s, a, s')$ (i.e., “the immediate reward obtaining when executing action a in state s and the system evolving to the next state s' ”).

The R&N book uses $R(s)$, but the more standard literature on Machine Learning uses $R(s, a, s')$. However, one can convert one to the other, possibly by adding additional states.

(a) Consider the following MDP representing the life of a student from David Silver’s class (check his [slides](#)):



Can you obtain the corresponding an alternative MDP that achieves the same modeling but with rewards per state, that is, with a reward function $R(s)$?

End of tutorial worksheet